Natural Language Processing

Lecture 16: Semantic Roles

11/30/22

COMS W4705
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Word Meaning and Sentence Meaning

- So far we have discussed the meaning of individual words.
- Today: meaning of entire predicate-argument structures and sentences.
- What should the representations be?
- How do we compute predicate or sentence-level representations from word representations?
  - What is the role of syntax?

Approaches to Sentence Level Semantics

- Semantic Role Labeling (SRL) / Frame Semantic Parsing.
  - Target representation: PropBank predicate argument structures, FrameNet-style annotations.
- Full-sentence semantics:
  - Target representations: Predicate-logic, Abstract Meaning Representation,...

Predicate-Argument Structures

- Verbs describe events or states (situations). "eventualities"

- Who does what to whom?

John$_{[subj]}$ ***gave*** *the book*$_{[obj]}$ *to Mary*$_{[pp\_to]}$
Giver **GivenObject** **Recipient**

- Grammatical Functions (subj, obj, ...)

- vs. semantic roles (Giver, GivenObject, ...)

Valency / Syntactic Frames

- Verbs may take a different number of **arguments** of different syntactic types in different positions
  - *The baby slept.* * *The baby slept the house.*
  - *He pretended to sleep.* **He pretended the cat.*
  - *Godzilla destroyed the city.* **Godzilla destroyed.*
  - *Jenny gave the book to Carl.* **Jenny gave the book.*

Diatheses Alternation

- The ability of a verb to appear in **a pair of syntactic frames (**"alternation")

- Examples:

Causative/Inchoative Alternation:

{break, chip, crack, crash, crush, fracture, rip, shatter, smash, snap, splinter, snip, tear}

John$_{\text{[subj]}}$ ***broke*** *the window*$_{\textit{[dobj]}}$ $\longrightarrow$ *the window*$_{\textit{[subj]}}$ ***broke***

John$_{\text{[subj]}}$ ***cut*** *the loaf*$_{\textit{[dobj]}}$ $\not\longrightarrow$ **the loaf*$_{\textit{[subj]}}$ ***cut***

Conative Alternation:

{hit, bang, bash, knock, hammer, pound, kick, tap, strike, ...}

John$_{\text{[subj]}}$ ***hit*** *the door*$_{\textit{[dobj]}}$ $\longrightarrow$ *John*$_{\text{[subj]}}$ ***hit*** *at the door*$_{\textit{[dobj]}}$

John$_{\text{[subj]}}$ ***broke*** *the door*$_{\textit{[dobj]}}$ $\not\longrightarrow$ **John*$_{\text{[subj]}}$ ***broke*** *at the door*$_{\textit{[dobj]}}$

More Alternation Examples

Middle Alternation:

John[subj] ***cut*** *the loaf[dobj]* → *the loaf[subj]* ***cut*** easily

John[subj] ***adored*** cats *[dobj]* ↛ **the cats[subj]* ***adored*** *easily*

Dative Alternation:

John[subj] ***gave*** *the book[obj]* *to Mary[pp_to]*

→ *John[subj]* ***gave*** *Mary[iobj]* *the book[obj]*

Locative Alternation:

John ***loaded*** *hay* *onto* *the wagon*

→ *John* ***loaded*** *the wagon* *with hay*

Alternations and Verb Meaning

- Observation:
  - Alternations preserve the meaning of the sentence.
  - The constituents change **grammatical function**, but continue to play the same **thematic role.**
  - Verb's participating **in the same set of alternations** seem to be semantically similar, that is they have some **common semantic component.**

John$_{[subj]}$ ***gave*** *the book$_{[obj]}$* *to Mary$_{[pp_to]}$*
Agent **Theme** **Recipient**

John$_{[subj]}$ ***gave*** *Mary$_{[iobj]}$* *the book$_{[obj]}$*
Agent **Recipient** **Theme**

give, deal, lend, loan, pass, peddle, refund, render

transfer-of-possession

Levin's Verb Classes

(Levin 1993)

- Group 3100 verbs, grouped into 47 verb classes (plus subclasses)

- Identify common semantic component.

break 45.1	{break, chip, crack, crash, crush, fracture, rip, shatter, smash, snap, splinter, snip, tear}	**change-of-state**
cut 21.1	{chip, chop, clip, cut, hack, hew, scratch, slash, ...}	**change-of-state, sharp-instrument**
hit 18.1	{hit, bang, bash, knock, hammer, pound, kick, tap, strike, ...}	**contact, exertion-of-force**
give 13.1	{give, deal, lend, loan, pass, peddle, refund, render}	**transfer-of-possession**

VerbNet

(Kipper Schuler 2006)

- Broad-coverage lexicon of verbs, based on a refinement of Levin's classes. Over 1000 hierarchically organized classes and subclasses, about 4000 verbs.
- Each verb is linked to a set of classes it appears in. Class members correspond to *coarse senses* of the verb.
- Classes contain:
  - a subset of **thematic roles**.
  - a **list of syntactic frames** (i.e. an alternation pattern), illustrating how thematic roles can be mapped to grammatical function.
  - a **semantic representation (*semantic predicates*)** specifying how the thematic roles interact.

VerbNet Example

(Kipper Schuler 2006)

No Comments

give-13.1

Members: 7, Frames: 4

POST COMMENT

CLASS HIERARCHY

- GIVE-13.1
  - GIVE-13.1-1

MEMBERS

KEY

DEAL (WN 5, 9, 11, 12; G 4)

LEND (WN 2; G 2)

LOAN (WN 1)

PASS (FN 1; WN 5, 20, 21, 22; G 4)

PEDDLE (WN 1; G 1)

REFUND (WN 1; G 1)

RENDER (WN 2, 6, 7, 8; G 2)

ROLES

REF

- AGENT [+ANIMATE | +ORGANIZATION]
- THEME
- RECIPIENT [+ANIMATE | +ORGANIZATION]

VerbNet Example

give 13.1

FRAMES REF KEY

NP V NP PP.RECIPIENT

EXAMPLE	"They lent a bicycle to me."
SYNTAX	AGENT V THEME {TO} RECIPIENT
SEMANTICS	HAS_POSSESSION(START(E), AGENT, THEME) HAS_POSSESSION(END(E), RECIPIENT, THEME) TRANSFER(DURING(E), THEME) CAUSE(AGENT, E)

NP V NP-DATIVE NP

EXAMPLE	"They lent me a bicycle."
SYNTAX	AGENT V RECIPIENT THEME
SEMANTICS	HAS_POSSESSION(START(E), AGENT, THEME) HAS_POSSESSION(END(E), RECIPIENT, THEME) TRANSFER(DURING(E), THEME) CAUSE(AGENT, E)

- The inventory of thematic roles and predicates is fixed.
- The semantic representation used is a kind of quantifier-free predicate logic, referring to an event variable E
(this is a kind of "neo-Davidsonian" event semantics, more later)

Predicate-Argument Structures

- Verbs describe events or states (situations). "eventualities"
- Who does what to whom?

John$_{[subj]}$ ***gave*** *the book*$_{[obj]}$ *to Mary*$_{[pp\_to]}$

Giver **GivenObject** **Recipient**

- Grammatical Functions (subj, obj, ...)
- vs. semantic roles (Giver, GivenObject, ...)

Fillmore's "Case for Case"

(Fillmore 1968)

- "deep cases" (later also known thematic roles or theta roles) are linguistic universals.
- Each verb requires a subset of these deep cases.
- Different languages use different surface realizations for these cases (for example morphological marking, position, prepositions, ...), depending on the verb

John ruined the table	What John did to the table was ruin it.
John built the table	(?) What John did to the table was build it.

Fillmore's "Case for Case"

- More evidence for the existence of deep cases

John broke the table

The hammer broke the window

John broke the window with the hammer

- Cannot conjoin two phrases realizing the same role

*John and the hammer broke the window

- Each case can only be realized once

*The hammer broke the window with the chisel

Thematic Roles

- Difficult to come up with a specific inventory of roles that generalizes across all verbs.
- A widely agreed upon set:

Agent	Initiator of action, capable of volition
Patient	Affected by action, under goes change of state
Theme	Entity moving, or being “located”
Experiencer	Perceives action but not in control
Beneficiary	For whose benefit action is performed
Instrument	Intermediary/means used to perform an action
Location	Place of object or action
Source	Starting point
Goal	Ending point

Identify the thematic roles

[The ball] flew [into the outfield.]

[Jim] gave [the book] [to the professor.]

[Jim] saw [the dog]

[Laura] talked [to the class][about the bomb threats.]

[Laura] scolded [the class.]

[Bill] cut [his hair] [with a razor.]

[Gina] crashed [the car] [with a resounding boom.]

Problems with Thematic Roles

- Difficult to agree on a specific inventory of roles that generalizes across all verbs and argument types.

- Roles are often difficult to identify

The rock$_{[subj]}$ ***broke*** *the window$_{[obj]}$*

Agent or Instrument or theme?

John$_{[subj]}$ ***broke*** *the window$_{[obj]}$*

Patient or Theme?

The window$_{[subj]}$ ***broke***

John$_{[subj]}$ ***touched*** *the book$_{[obj]}$*

- Semantic properties of arguments provide clues:

  - John is +animate, rock is -animate

- Agent: *What did X do?* Patient: *What did X do to Y?*

Problems with Theta Roles (2)

- Chomsky's Theta criterion (Chomsky 1982):
  - one-to-one correspondence between noun phrases and thematic roles.
  - Counter-example:

Esau ***sold*** *his birthright to Jacob for a bowl of porridge*

Agent **Theme** **Goal** **??Theme??**

- Solutions:
  Fewer thematic roles (Dowty's proto-roles [Dowty 1991])
  More thematic roles (Frame semantics [Fillmore 1986],
  FrameNet: different role inventory
  for each frame)

Proto Roles

(Dowty 1991)

- Role types are not discrete categories, but "cluster concepts" (as in Rosch's prototype theory)
- Proto-Agent
  - Volitional involvement in event or state.
  - Sentience (and/or perception)
  - Causes an event or change of state in another participant
  - Movement (relative to position of another participant)
- Proto-Patient
  - Undergoes change of state
  - Causally affected by another participant
  - Stationary relative to movement of another participant

PropBank

(Palmer et al. 2005)

- Goals:
    - Large-scale semantic annotations (predicate/argument structures) over syntactically annotated text (most verbs).
    - Consistent labeling of arguments for each subcategorization frame of a verb. "Functional tags" (ArgM) for modifiers/adjuncts.
- Unlike VerbNet (or FrameNet): No grouping of verbs into classes (but see SemLink and the Unified Verb Index).
- Source data: Originally Penn Treebank, now Ontonotes 5.0. Large variety of corpora in other domains (clinical notes, image captions, web treebank,...)
- More recent versions cover nouns (Hwang et al., 2010) and predicative adjectives (Bonial et al., 2017)
- Versions in other languages

PropBank Roles

- Roles are simply numbered (Arg0, Arg1, ..., ArgM)
  Interpretation of roles is verb-sense specific, but as consistent as possible.

the company	donated	over $35,000	to residents
Arg0	**rel**	Arg1	Arg2

- Arg0:PROTO-AGENT
- Arg1:PROTO-PATIENT
- Arg2: usually: benefactive, instrument, attribute, or end state
- Arg3: usually: start point, benefactive, instrument, or attribute
- Arg4 the end point

PropBank ArgM and modifiers

- ArgM are used to annotated modifiers/adjunct-like phrases.
  These are consistent between verbs.
  (some modifiers are also used for roles)

- COM: Comitative
- LOC: Locative
- DIR: Directional
- GOL: Goal
- MNR: Manner
- TMP: Temporal
- EXT: Extent
- REC: Reciprocals
- PRD: Secondary Predication
- PRP: Purpose
- CAU: Cause
- DIS: Discourse
- ADV: Adverbials
- ADJ: Adjectival
- MOD: Modal
- NEG: Negation
- DSP: Direct Speech
- LVB: Light Verb
- CXN: Construction

PropBank Frame sets

- Each verb-sense corresponds to a **frame-set**, which specifies
  - a set of roles and their definition **(role-set,** ~theta grid)
  - a set of subcategorization frames
- All frame-sets for a verb form a **frames file**

Frameset for tend.01, *care for*

Arg0: tender
Arg1: thing tended (to)

	John Arg0	tends **rel**	to the needs of his patrons Arg1

Frameset for tend.02, *have a tendency*

Arg0: theme
Arg2: attribute

	The premium Arg0	tends **rel**	to get fat Arg2	in times of crisis ArgM-TMP

Another Example

Frameset for increase.01, *go up incrementally*

Arg0: causer of increase
Arg1: thing increasing
Arg2: amount increased by
Arg3: start point
Arg4: end point

[Arg0 Big Fruit Co.] **increased** [Arg1 the price of bananas]

[Arg1 The price of bananas] was **increased** again [Arg0 by Big Fruit Co.]

[Arg1 The price of bananas] **increased** [Arg2 by 5%]

Observations:

Syntax and semantics do not map 1:1. Generalize away from syntactic variations.

PropBank senses are coarse (for example, metaphorical uses treated the same as literal senses).

Frame Semantics

(Fillmore, 1982)

- Long history in cognitive science, AI, ... **(Minksy 1974, Barsalou 1992, Löbner 1998)**
- A **frame** describes the entire conceptual structure ("cognitive/ knowledge schema") needed to understand a word.
- Different words (of different part-of-speech, even phrases) can evoke the same frame

TRANSFER → *{give.v, receive.v, donate.v, gift.n, ...}*

CHANGE_OF_LEADERSHIP → *{coronate.v, coup.n, crown.v, depose.v, dethrone.v, elect.v, election.n, enthrone.v, install.v, oust.v, ..., revolution.n,...}*

Additional Issues with Thematic Roles / Deep Cases

"*I finally* ***replaced*** *the bicycle that was stolen*"

- Typically, thematic roles are thought of as event participants.
- What is the thematic role for "the bicycle that was stolen"?
- Problem: The bicycle that was stolen is not actually part of the event. It is certainly not the **theme**.

Example from Fillmore's ACL Lifetime Achievement Award Keynote

Additional Issues with Thematic Roles / Deep Cases (2)

- (1) The teacher **gave** the student a book.
  Agent **Recipient** **Theme**

- (2) The student **received** a book from the teacher.
  Recipient **Theme** **Source**

- (1) and (2) seem to describe the same event. It seems odd that "the teacher" is a Source with **receive** and an Agent with **give.**

- The true role of "the teacher" in this event seems to combine properties of both Source and Agent.

Frame Elements

- Frames describe the interaction/relation between a set of frame-specific semantic roles called Frame Elements (FEs).
- Each word in a frame (lexical unit LU) has its own lexical entry listing possible valence patterns (i.e. linkings between frame elements and grammatical functions).

TRANSFER

This frame involves a Donor transferring a Theme to a Recipient.

give.v

Donor	Theme	Recipient
subj	iobj	obj

Donor	Theme	Recipient
subj	obj	PP-to

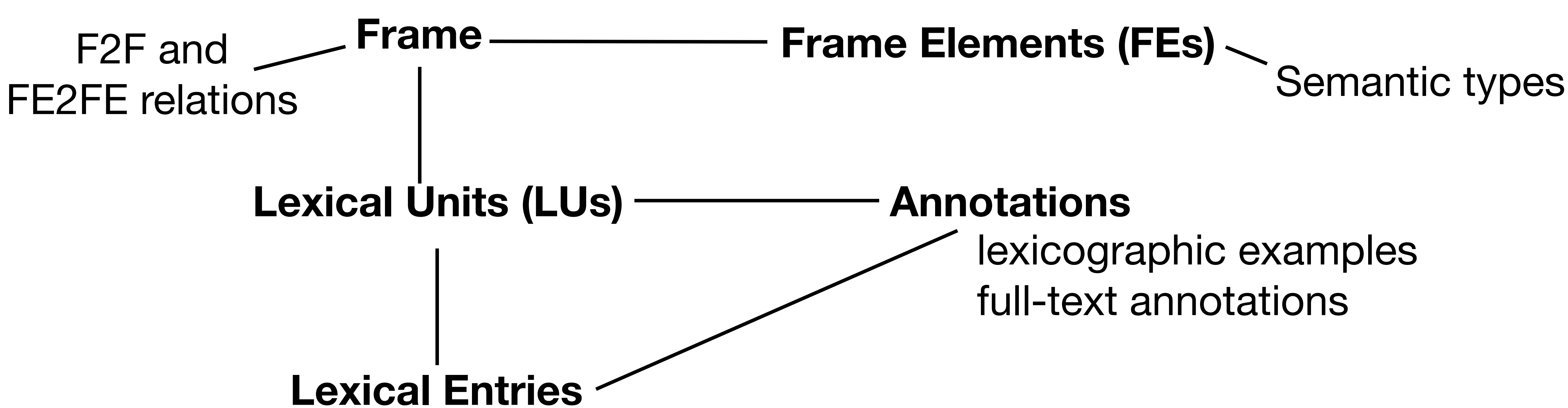
receive.v

Donor	Theme	Recipient
pp-from	obj	subj

FrameNet Overview

(Baker et al, 1998)

- Lexical resource based on Frame Semantics:
  13000+ lexical units in 1200+ frames.



Giving Frame

A Donor transfers a Theme from a Donor to a Recipient.

Inherits from: Intentionally_act, Lose_possession
Is Inherited by: Commerce_pay, Commerce_sell, Lending, Submitting_documents, Supply, Surrendering_possession
Perspective on: Transfer
Is Used by: Offering

Lexical Units: *{advance.v, bequeath.v, bequest.n, ..., give.v, gift.n, ...}*

Core:

Donor *The person that begins in possession of the Theme and causes it to be in the possession of the Recipient*

Recipient *The entity that ends up in possession of the Theme.*

Theme *The object that changes ownership.*

Non-core:

Means *The Means by which the Donor gives the Theme to the Recipient.*

Purpose The Purpose for which the Donor gives the Theme to the Recipient.

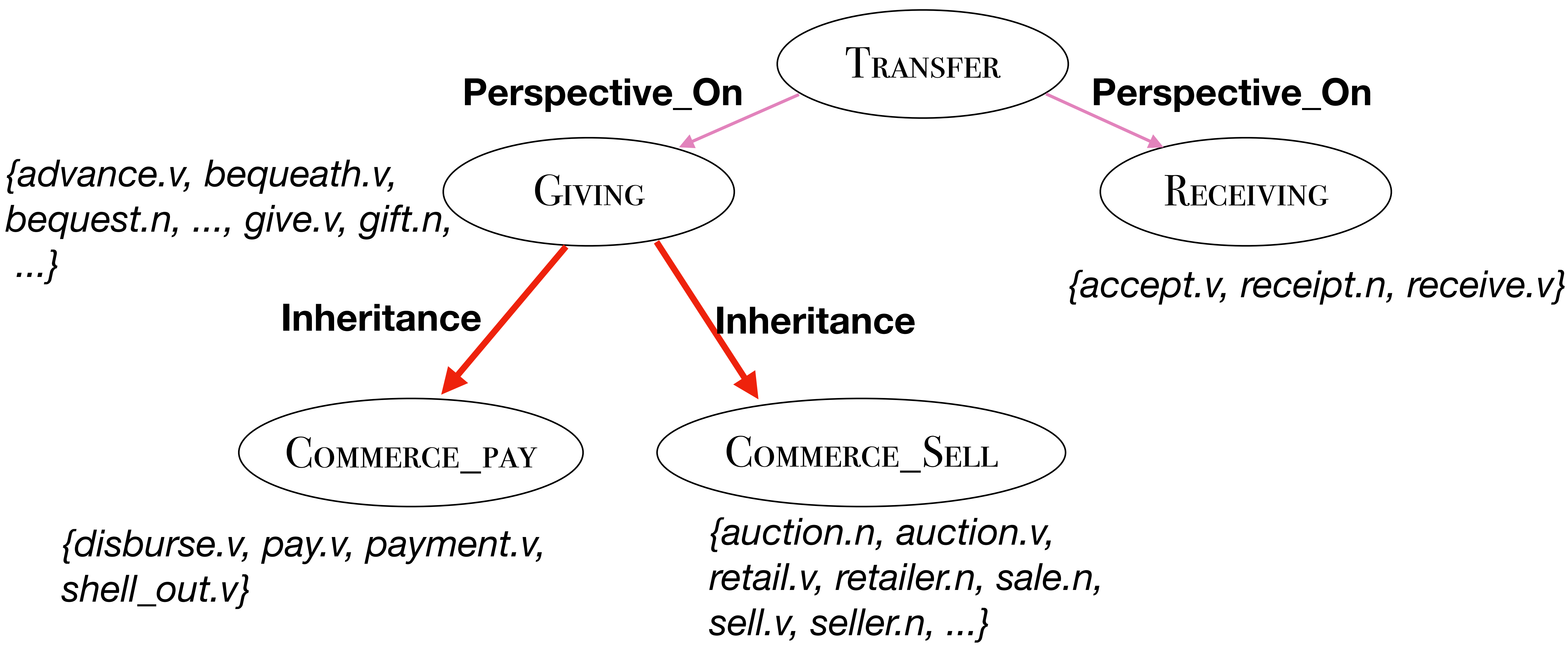
Place

Time

⋮

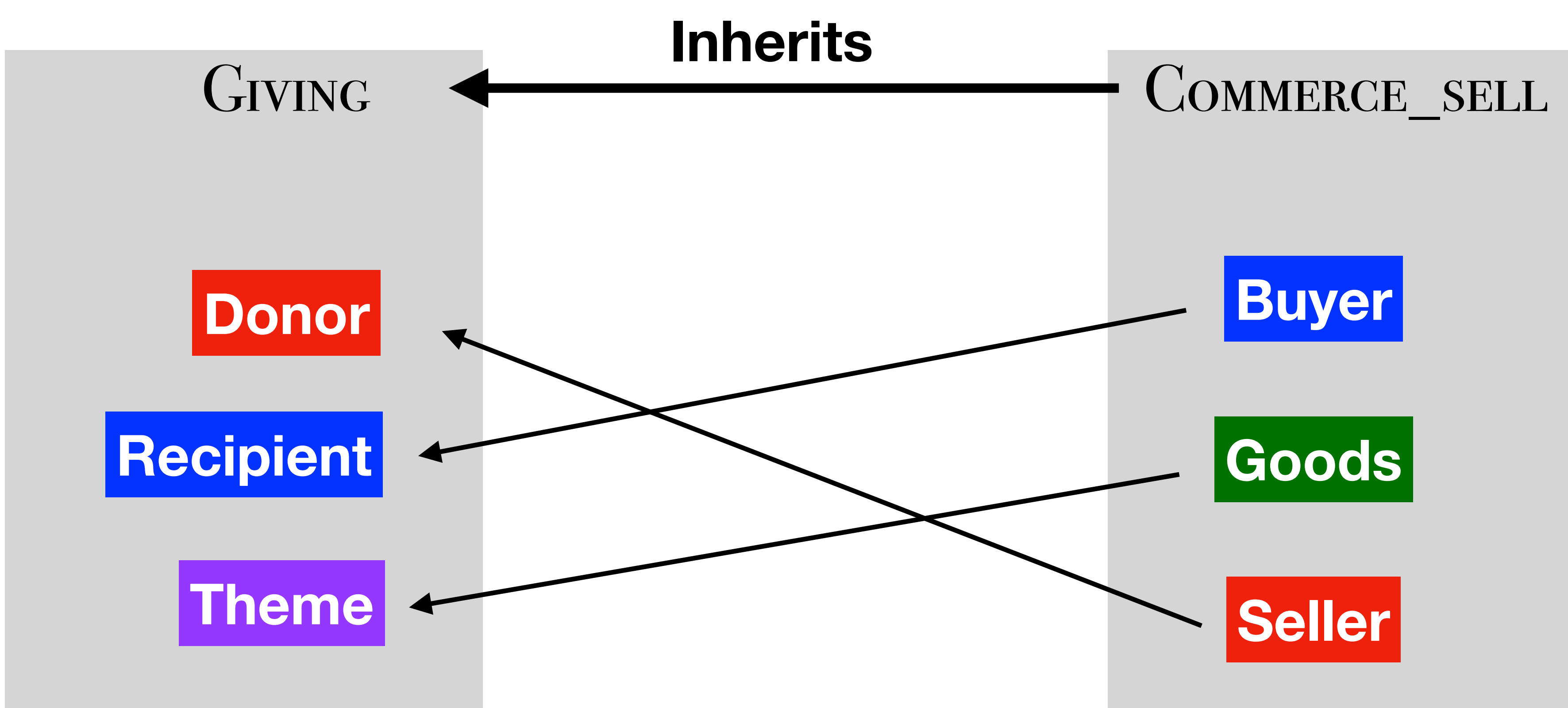
Relations between Frames

- Frames can be related to each other in various ways.



Frame-Element Relations

- Some frame-to-frame relations require mapping of frame elements.



Annotations and Lexical Entries

- **Annotations** illustrate how frame elements are realized linguistically (linkings/ valence patterns).

- There are two types: Lexicographic examples and fulltext-annotations (several "corpora", different genres, including some PropBank WSJ docs)

	Apple	wanted to	**donate**	a computer	to every school in the country	.
POS	NNP	VVD TO	VB	DT NN	PRP DT NN IN DT NN	.
FE	**Donor**		**FEE**	**Theme**	**Receipient**	
GF	Subj			Obj	Dep-to	
PT	NP			NP	PPto	

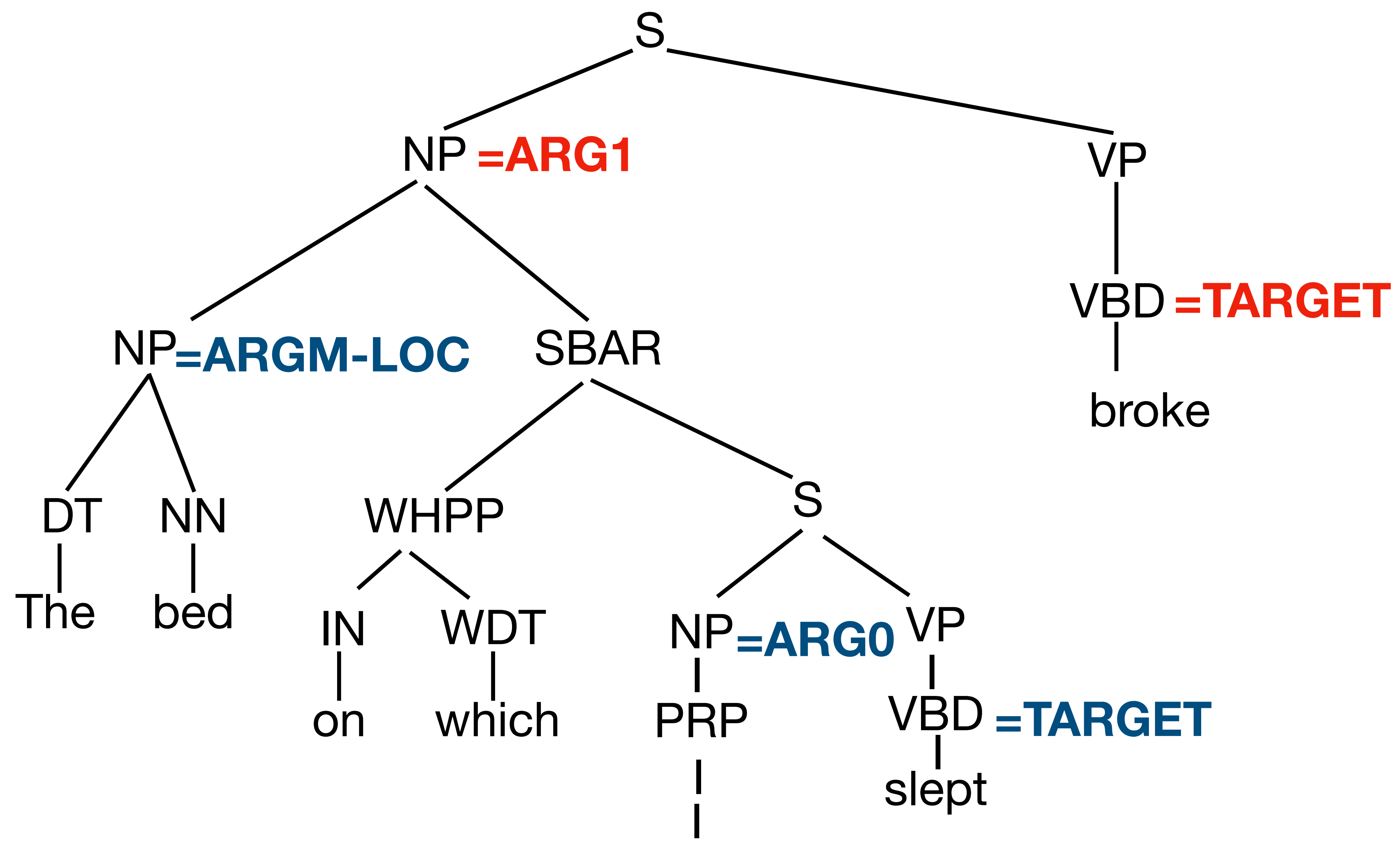
- **Lexical Entries** summarize different LUs for each word, together with possible realization (grammatical function) for each FE.

Semantic Role Labeling

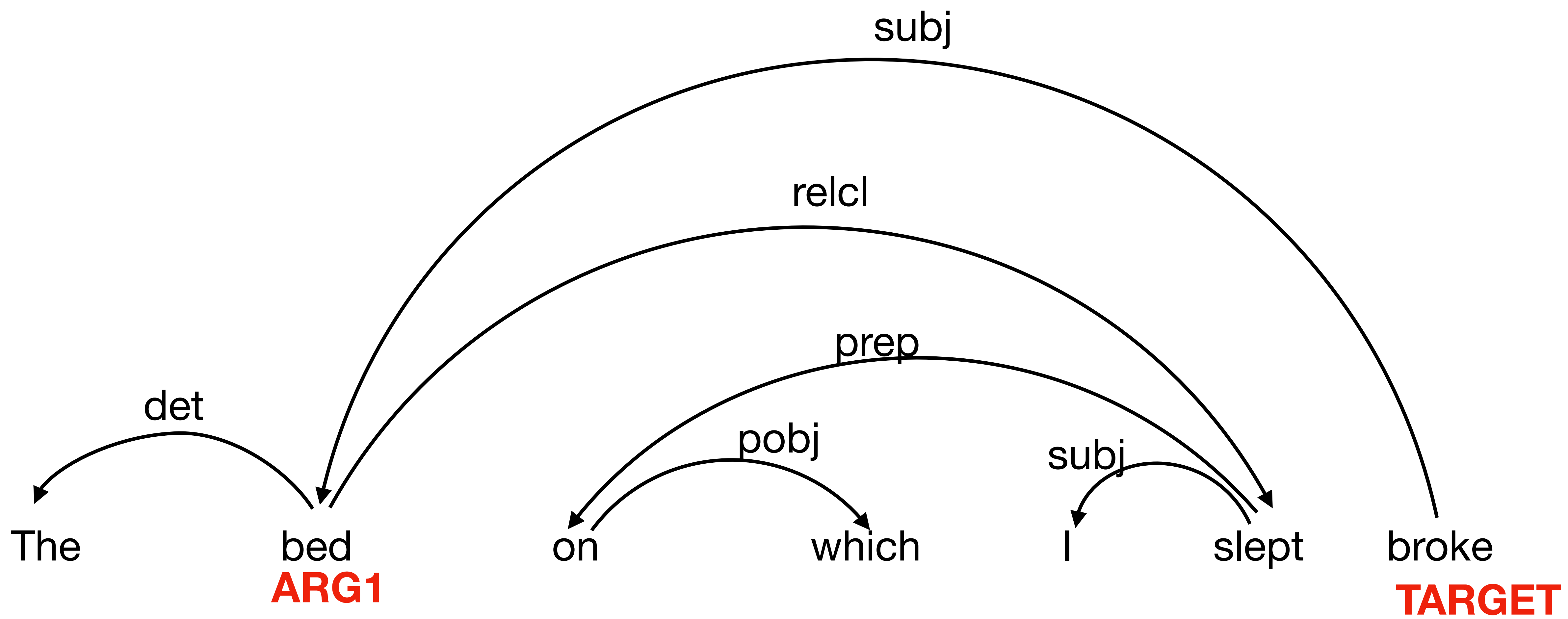
Generic Syntax-Based SRL Algorithm

- Parse the sentence (dependence or constituency parse)
- Detect all potential targets (predicates / frame evoking elements) **1. target identification**
- For each target:
  - For each node in the parse tree, use supervised ML classifiers to:
    - **2. argument identification**
    - **3. argument labeling**

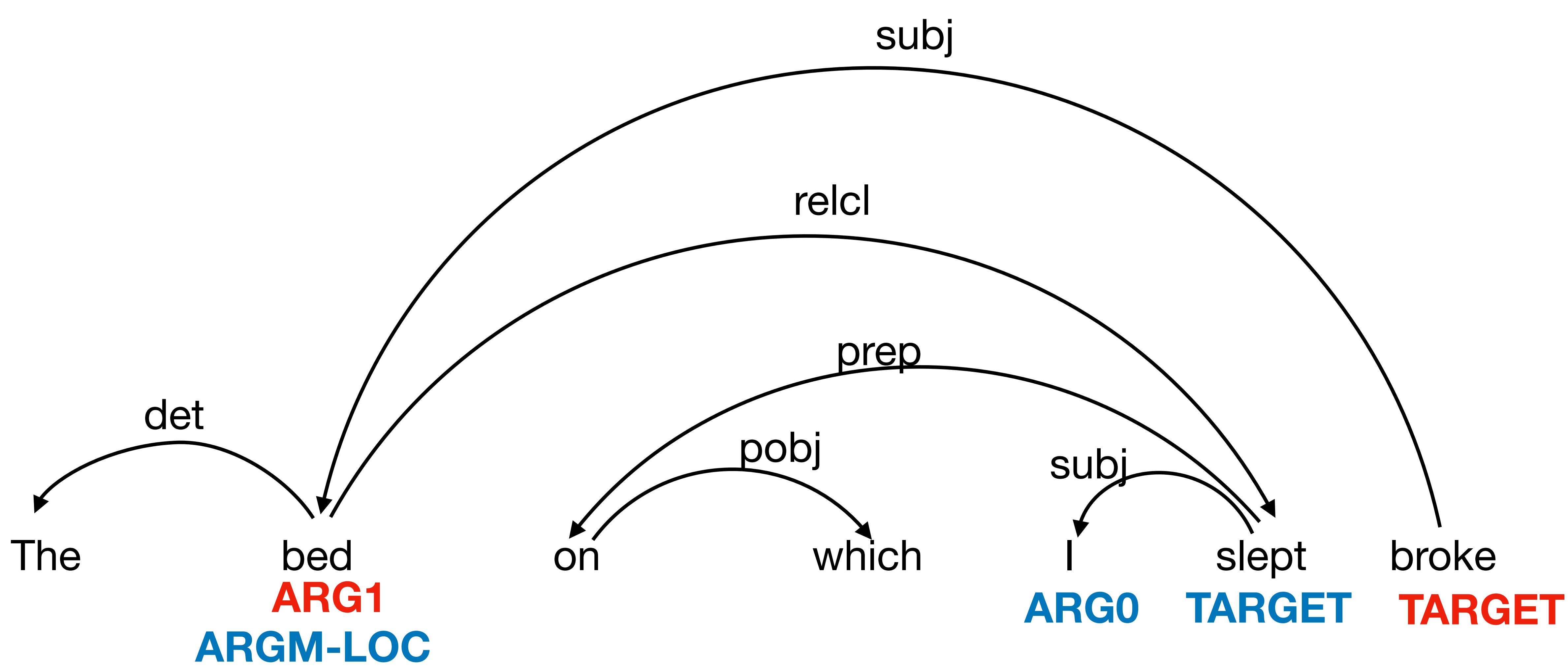
SRL Example (PropBank)



SRL Example (PropBank)



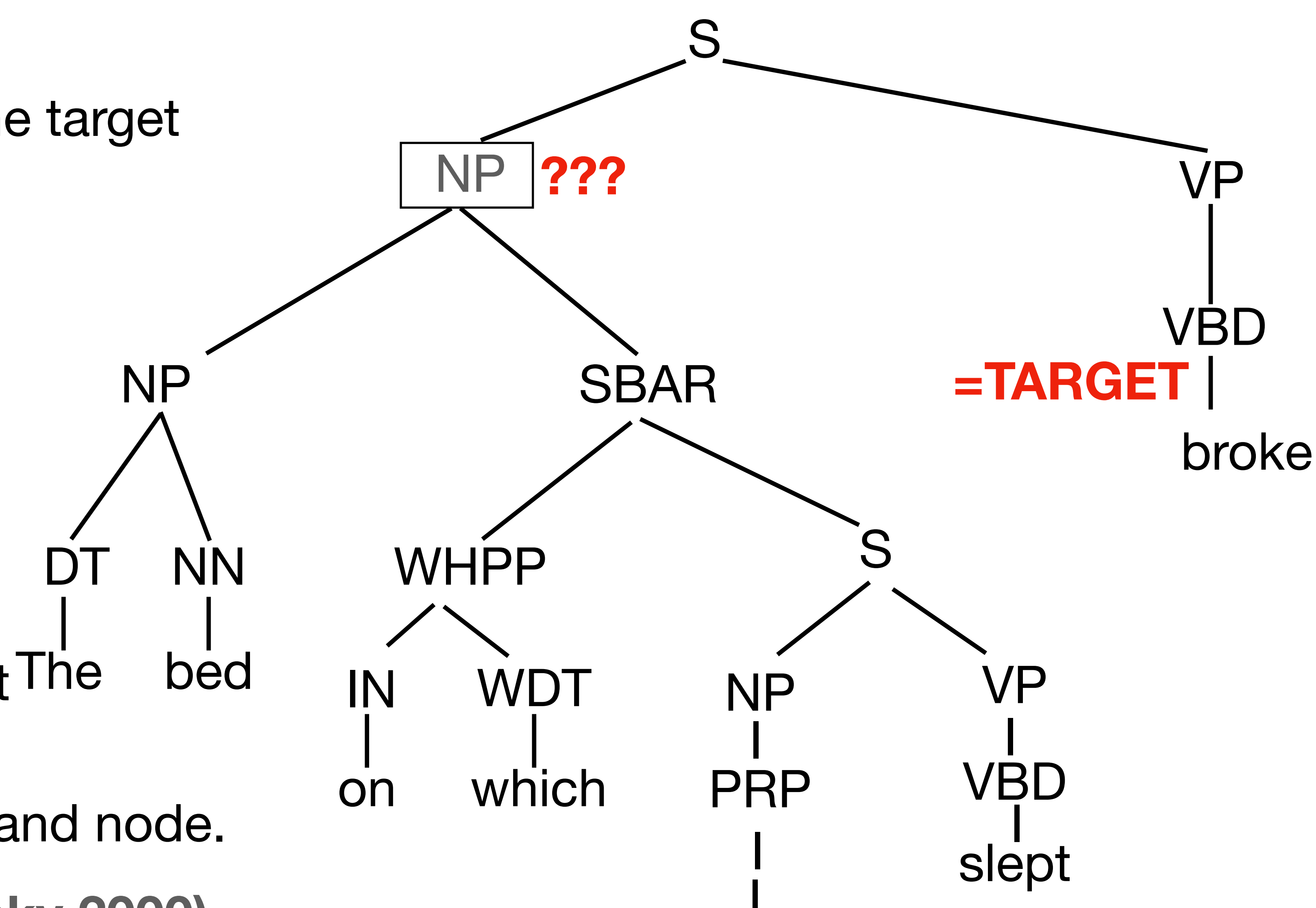
SRL Example (PropBank)



Features

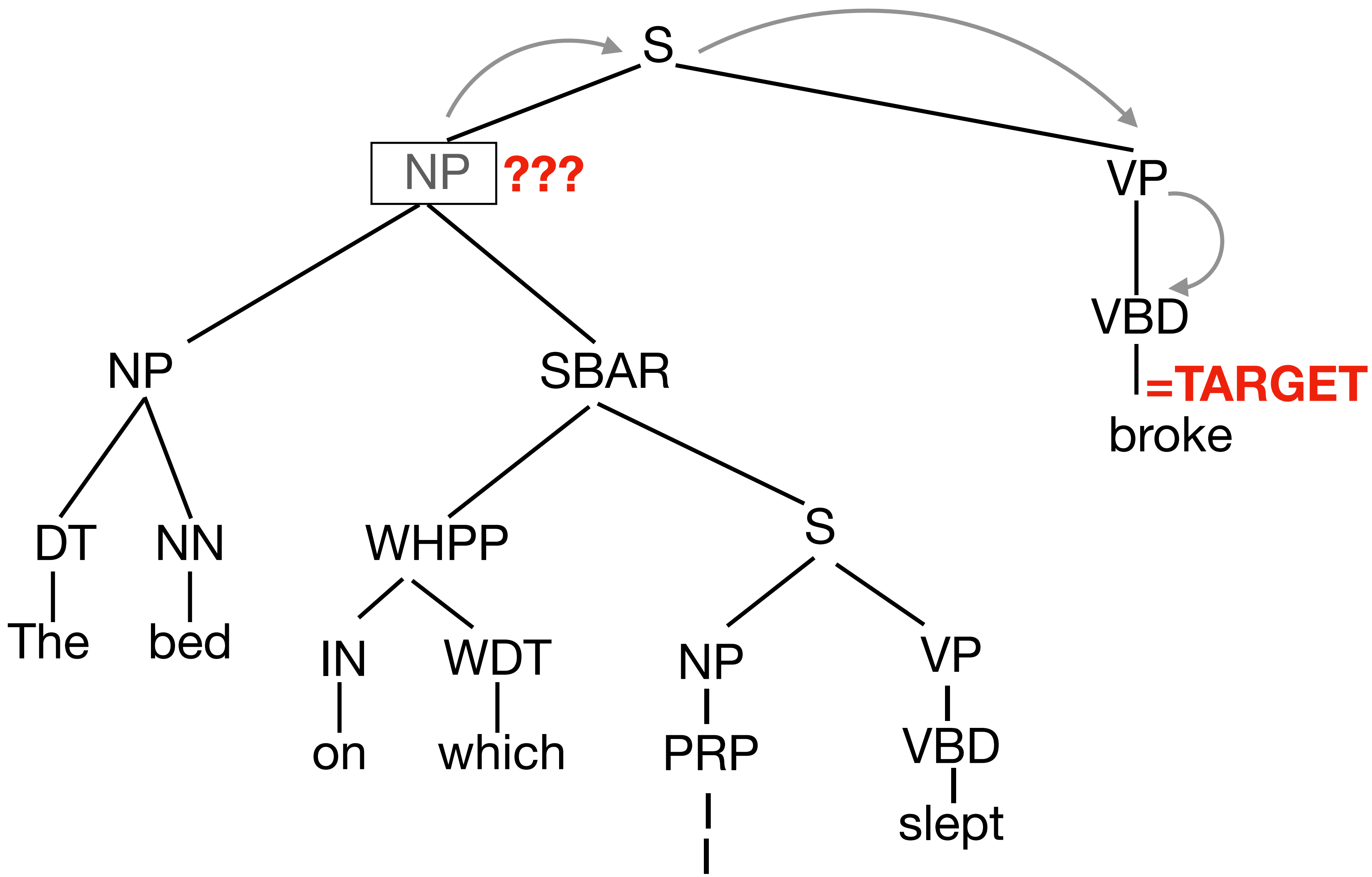
What features should we use for argument detection and labeling?

- target predicate: broke
- headword (+POS): *bed NN*
- phrase type: NP
- linear position: before or after the target
- argument structure of the verb. "NP broke"
- target voice: active
- possibly semantic features (named entity class, WordNet synsets of head word, ...)
- first and last word of constituent and their POS.
- Parse tree path between target and node.



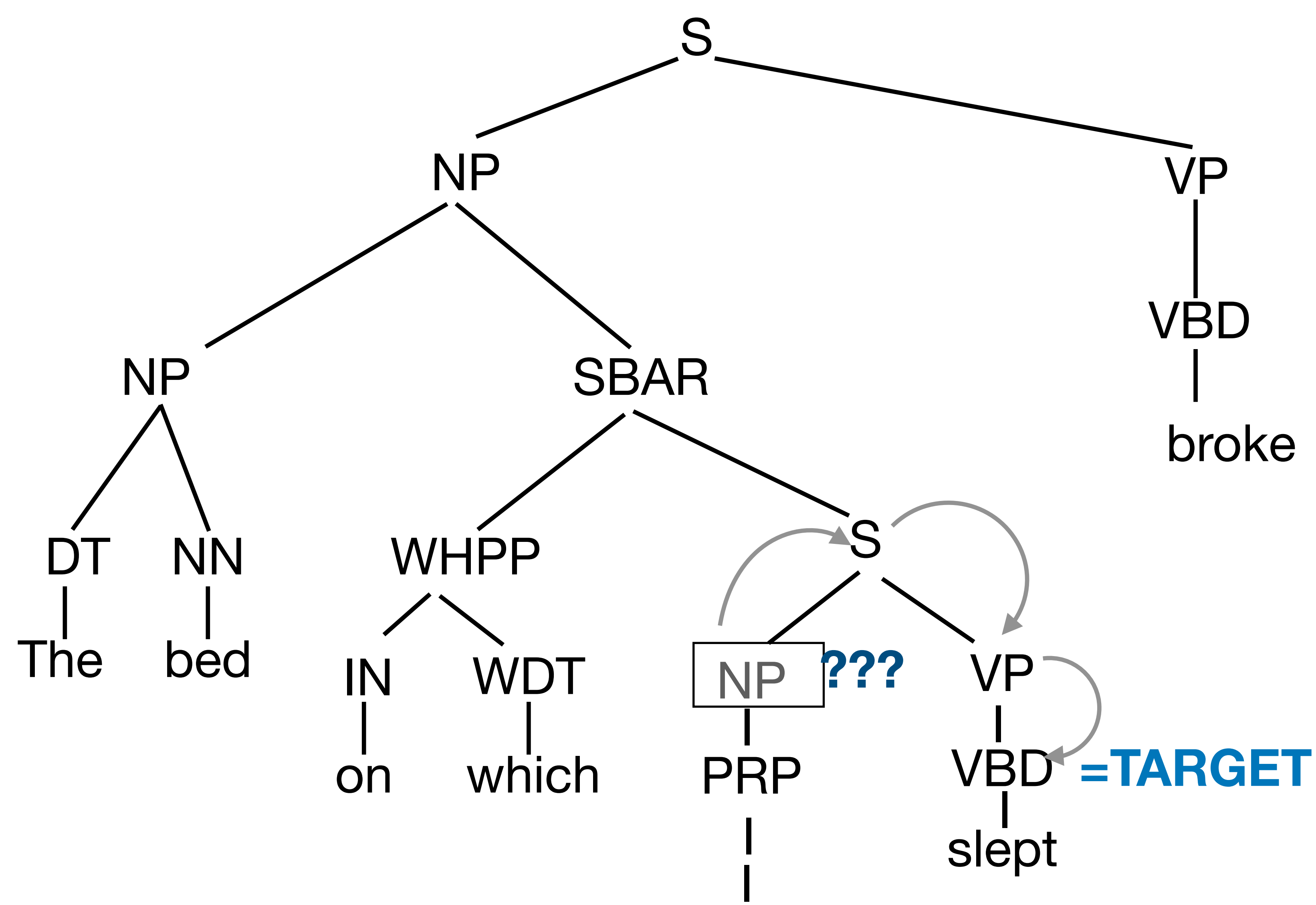
(Features used in Gildea & Jurafsky 2000)

Parse Tree Path



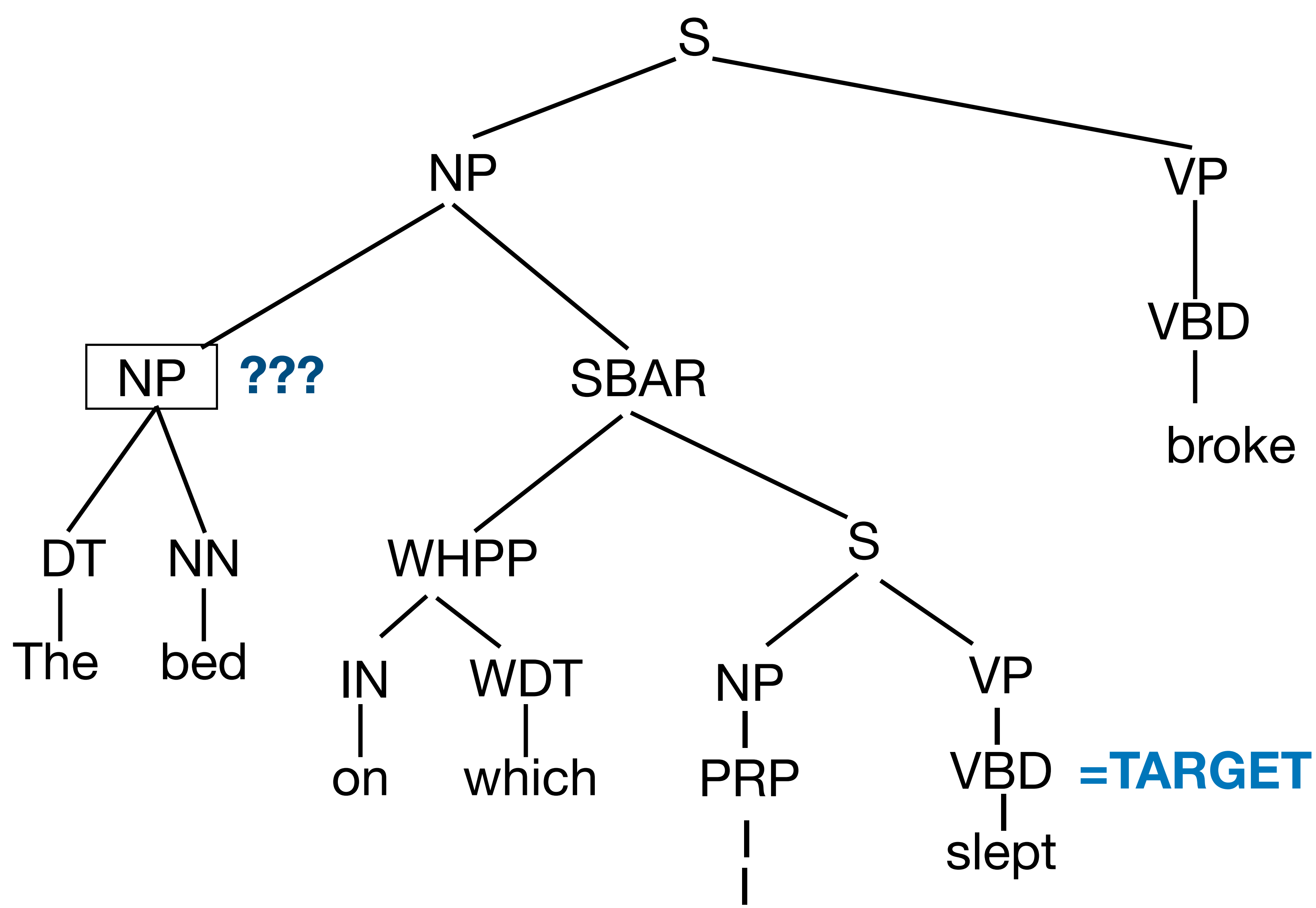
NP↑S↓VP↓VBD

Parse Tree Path



NP↑S↓VP↓VBD

Parse Tree Path

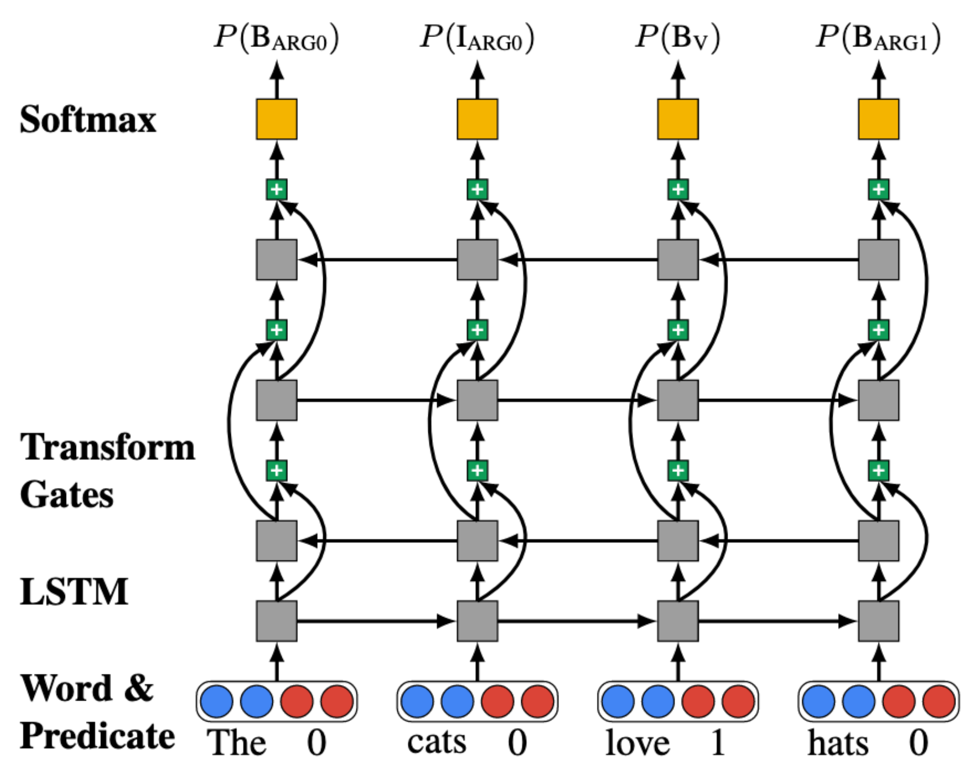


NP↑NP↓SBAR↓S↓VP↓VBD

RNN based SRL

(Collobert et al. 2011, Zhou & Xu, 2015, He et al. 2017)

- Treat SRL task as a sequence labeling task.
  - Input is a sequence of tokens in the sentence and a binary marker indicating the predicate.
  - Output is a sequence of B I O tags (for each ARG*N*).



Highway LSTM for SRL

More recent work uses transformers, e.g. Strubbel et al. 2018

RNN based SRL - Constraints

(He et al. 2017)

- Enforce constraints on the output, such as
  - Result must be a valid B I O sequence of matching ARGs (for example, don't allow B_{ARG0} followed by I_{ARG1})
  - Each role can appear only once.
- Add a penalization term to the scoring function:

$$f(\boldsymbol{w}, \boldsymbol{y}) = \sum_{t=1}^{n} \log p(y_t \mid \boldsymbol{w}) - \sum_{c \in \mathcal{C}} c(\boldsymbol{w}, y_{1:t})$$

  - where $\sum_{c \in \mathcal{C}} c(\boldsymbol{w}, y_{1:t})$ is the sum of non-negative penalties imposed by the constraints over the output sequence up to time t.
- Find highest scoring sequence using A* decoding.

RNN based SRL - Decoding

(He et al. 2017)

- Goal: Using the log-probabilities produced by the RNN, find the highest scoring output that satisfies the constraints.
- Built the output sequence left to right and perform an A* search over the prefixes of the sequence.
- Score for each prefix: $f(\boldsymbol{w}, y_{1:t}) = \sum_{i=1}^{t} \log p(y_i \mid \boldsymbol{w}) - \sum_{c \in \mathcal{C}} c(\boldsymbol{w}, y_{1:i})$
- Admissible A* heuristic: sum max probability for each token to the right of t.

$$g(\boldsymbol{w}, y_{1:t}) = \sum_{i=t+1}^{n} \max_{y_i \in \boldsymbol{T}} \log p(y_i \mid \boldsymbol{w})$$

- Explore prefixes in the order determined by

$$f(\boldsymbol{w}, y_{1:t}) + g(\boldsymbol{w}, y_{1:t})$$

Span-graph based SRL

(He et al. 2018)

- Instead of BIO tagging, make independent predictions for each possible pair of predicate and argument span.

- Consider all words as possible predicates. $\mathcal{P} = \{w_1, \ldots, w_n\}$

- Consider all possible subspans of the sentence

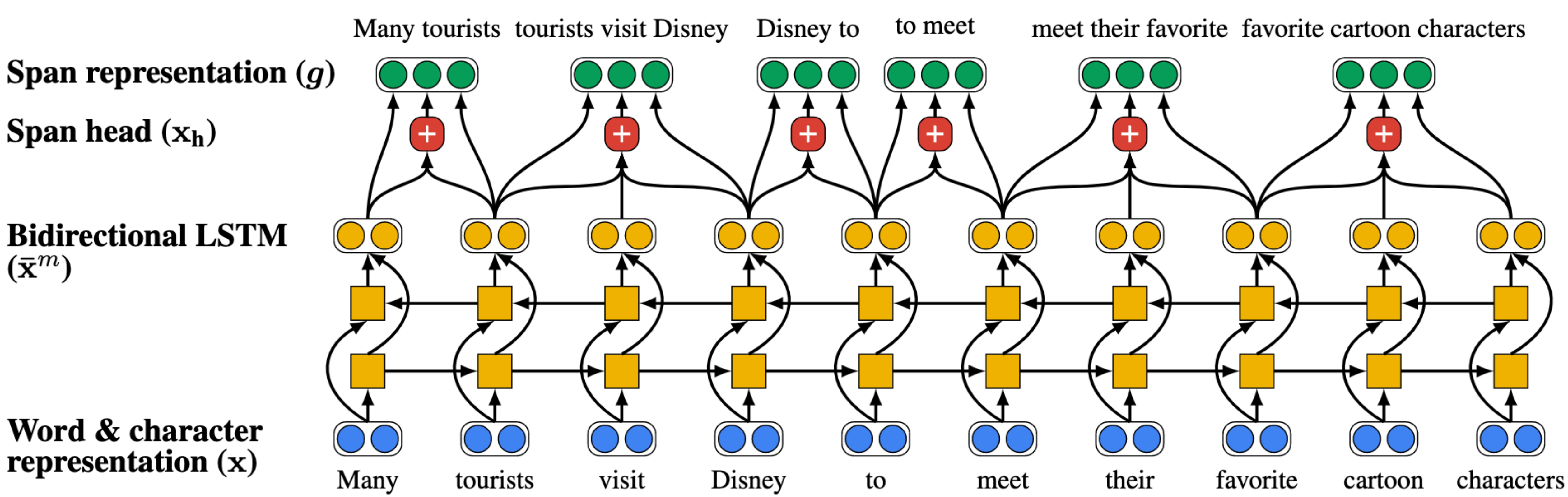
$$\mathcal{A} = \{(w_i, \ldots, w_j) | 1 \leq i \leq j \leq n\}$$

Span-graph based SRL Contd.

(He et al. 2018)

- Use a BiLSTM model to compute token representations. Then combine token representations into span representations.

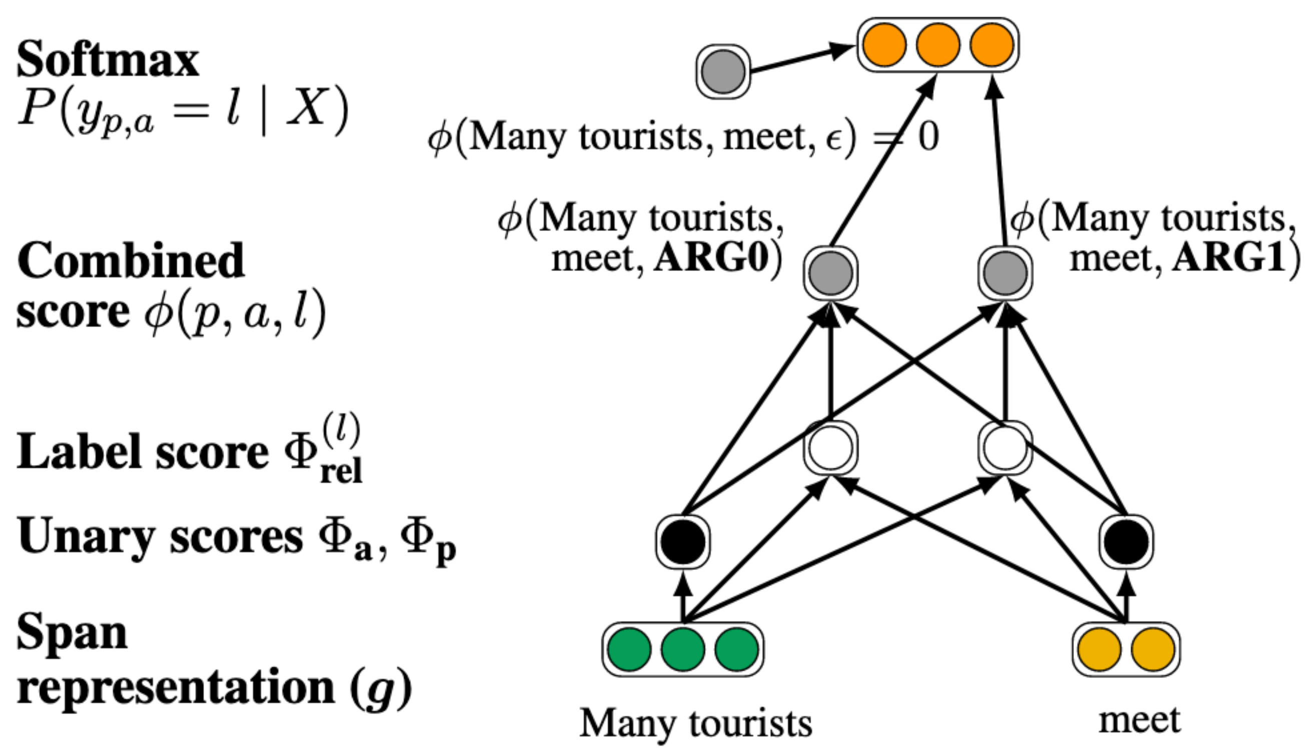
  - concatenate start word, end word, and an attention based representation of the span's head.



Span-graph based SRL Predictions

(He et al. 2018)

- Finally predict a label for the predicate / argument combination.



PropBank SRL Results

	CoNLL 2012 F1 score (gold predicates)	CoNLL 2012 F1 (predicated predicates)
Täckström et al. 2015 (syntactic)	79.4	70.3
He et al. 2017	81.7	-
He et al. 2018	82.1	82.9
He et al. 2018 + ELMo	85.5	79.8